

Exp 6 Finite Word length Effects

6.1 Uniform Quantization and Quantization Error Analysis of a Discrete-Time

Sinusoidal Signal Using MATLAB

Program:

```
clc;
clear;
close all;

%% Signal
fs = 1000;
f = 50;
t = 0:1/fs:0.1;
x = sin(2*pi*f*t);

%% Quantization Levels
L = [8 16 32];
colors = {'r','g','m'};
figure;
for i = 1:length(L)
    % Quantization
    xq{i} = round(((x + 1) * (L(i)-1) / 2)) * (2/(L(i)-1)) - 1;

    % Error & MSE
    e{i} = x - xq{i};
    mse(i) = mean(e{i}.^2);

    % Plot Quantized Signals
    subplot(4,1,i+1);
    stairs(t, xq{i}, colors{i}, 'LineWidth', 1.2);
    title(['Quantized Signal (' num2str(L(i)) ' Levels)']);
    xlabel('Time (s)');
    ylabel('Amplitude');
```

```

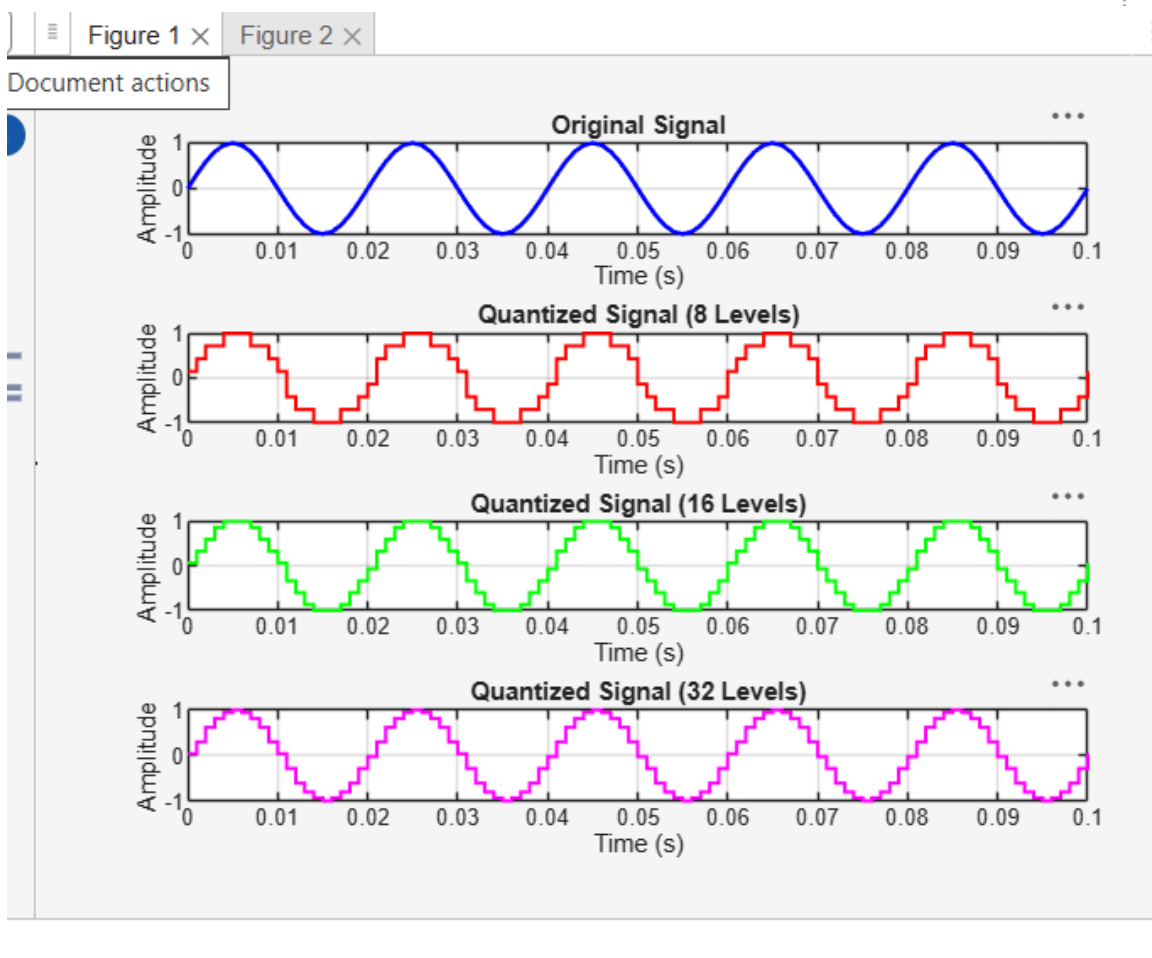
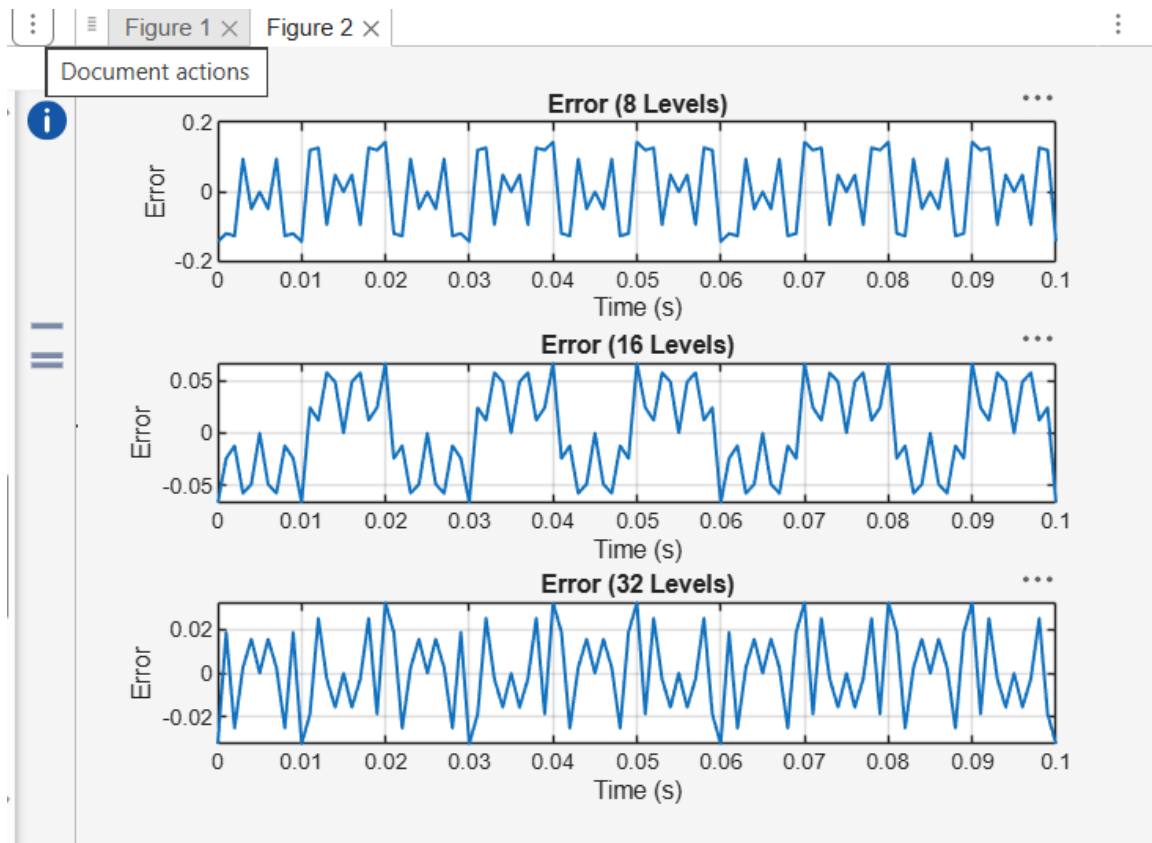
        grid on;
    end

    %% Original Signal
    subplot(4,1,1);
    plot(t, x, 'b', 'LineWidth', 1.5);
    title('Original Signal');
    xlabel('Time (s)');
    ylabel('Amplitude');
    grid on;

    %% Display MSE
    fprintf('MSE for 8 Levels = %f\n', mse(1));
    fprintf('MSE for 16 Levels = %f\n', mse(2));
    fprintf('MSE for 32 Levels = %f\n', mse(3));

    %% Error Plots
    figure;
    for i = 1:3
        subplot(3,1,i);
        plot(t, e{i}, 'LineWidth', 1.2);
        title(['Error (' num2str(L(i)) ' Levels)']);
        xlabel('Time (s)');
        ylabel('Error');
        grid on;
    end
end

```



6.2 Analysis of Finite Word Length Effects in Digital Signal Representation Using Truncation and Rounding Techniques in MATLAB

Program:

```
clc; clear; close all

n=0:79;

x=sin(2*pi*n/40);

q=8;

xt=floor(x*q)/q;

xr=round(x*q)/q;

et=x-xt;

er=x-xr;

% Fig 1

figure

plot(n,x,'b',n,xr,'r','LineWidth',1.5)

grid on

title('Original vs Quantized')

legend('Original','Quantized')

% Fig 2

figure

plot(n,x,'b',n,xt,'r',n,xr,'g','LineWidth',1.5)

grid on

title('Truncation vs Rounding')

legend('Original','Truncated','Rounded')

% Fig 3

figure

stem(n,et,'r')

hold on

stem(n,er,'g')
```

```
grid on
```

```
title('Quantization Error')
```

```
legend('Truncation','Rounding')
```

```
% Fig 4
```

```
figure
```

```
bar([mean(et.^2) mean(er.^2)])
```

```
set(gca,'XTickLabel',{'Truncation','Rounding'})
```

```
grid on
```

```
title('MSE Comparison')
```

```
ylabel('MSE')
```

